

Lab Activity

Assume the role of a junior software developer assigned to overhaul a legacy codebase named `student-portal/`.

Execution Objectives

Execute **9 practical tasks** independently while maintaining clean, professional, and meaningful Git history.

Focus on operational understanding rather than copying commands.



Lab Roadmap: Tasks 1 to 5

Parts 1–3: Init & Feature Building

Part 1 : Create local repo, set .gitignore, perform initial commit, and link GitHub remote. **[2 Marks]**

Part 2 : Enhance README.md with setup instructions (Rule: Min 6 separate commits). **[3 Marks]**

Part 3 : Build feature branch, create code files (e.g., main.cpp), and merge to main. **[5 Marks]**

Parts 4–5: File Ops & Scale

Part 4 : Rename, move, delete, and restore files while preserving full history lineage. **[10 Marks]**

Part 5 : Build 4 feature branches (login, profile, dashboard, settings) with 4-5 commits each and merge all. **[12 Marks]**

Lab Roadmap: Tasks 6 to 9

Parts 6–9 : Employing Git Tools

Part 6 : Create unfinished changes, stash them, switch to another branch, return to the original branch, and reapply the stashed changes. [5 Marks]

Part 7 : Resolve 3 specific mistakes (wrong message, misplaced commits, reflog recovery). [2 + 2 + 2 Marks]

Part 8 & 9 : Manually resolve created branch conflicts and release annotated tag v1.0 [7 Marks]

Lab Rules & Deliverables

☰ Strict Requirements:

- Avoid empty, trivial, or placeholder implementations in source code files. [-5 Marks]
- Minimum **40 total commits** and minimum **6 branches** created. [-15 Marks each]
- Descriptive commit messages only (no generic "fix" or "update"). Zero force-pushing. [-5 Marks]
- Deliverables: GitHub Repository URL + Screenshots of git log --graph --oneline --all [-50 Marks]